Paper 4: Splitting of Reciprocal Dual Cells and Hierarchical State Structure

A Minimal Observational Model of Internal State Capacity, Volume Gap, and Duality Breaking from $\nu\lambda = 1$

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Abstract

In this paper, starting from the minimal reciprocal dual condition placed between wavelength components λ_n and frequency components ν_n,

$$\lambda_n \nu_n = 1,$$

we reconstruct, as a single minimal observational model, the wavelength space, the frequency space, the 4-dimensional lattice cell counting, and the closed 4-dimensional structure treated in the three preceding observational papers.

The purpose of this paper is not to replace standard physical theory, general relativity, quantum theory, particle physics, or cosmology. Nor does this paper claim to have derived spacetime, mass, energy, interactions, the Standard Model, quantum gravity, or a grand unified theory. The λ_n and ν_n treated in this paper are model variables for observing the dual geometric structure of wavelength space and frequency space.

In this paper we show that, when the possible existence of a minimal conjugate width $\delta_{\min}$ is added to the reciprocal dual condition, a state is treated not as a mathematical point but as a conjugate cell with a minimal width. In this case, the composite frequency radius

$$R^2 = \sum_n \nu_n^2$$

can be read not as a substantive radius given from outside, but as an energy-like scale determined from the internal frequency components. $R = 1$ is observed as the minimal conjugate cell, $R = 2$ as eight first-neighbor quasi-ground states, and $R = 3$ as a large internal state shell containing 128 additional states.

Furthermore, by comparing the 4-dimensional hyperball volume

$$V_4(R) = \frac{\pi^2}{2} R^4$$

with the fully-inscribed cell volume $N_0(R)$, we confirm that the filling fraction shows an asymptotically increasing tendency accompanied by local fluctuations, while the volume gap as an absolute quantity,

$$\Delta V(R) = V_4(R) - N_0(R),$$

grows. From this observation, we introduce a toy model that reads a single state with high R as a decomposition into many finite-R' states.

Finally, we describe an observation of duality breaking: in the initial state the roles of ν and λ are undifferentiated, and through the splitting into finite-R' cells and the minimal conjugate width, the frequency side differentiates into dense internal states while the wavelength side differentiates into a sparse extension. This observation does not derive the differentiation of time and space. However, it does serve as a toy model in which the ν side is read as internal updating and state density, and the λ side as extension and spread.

The conclusion of this paper is that, from the minimal reciprocal dual assumption $\nu\lambda = 1$, the minimal conjugate cell, internal state capacity, volume gap, finite-R' decomposition, duality breaking, and curved hierarchical structure can be read in a chained manner within the same observational model.

Keywords

wavelength space, frequency space, reciprocal duality, $\nu\lambda = 1$, conjugate cell, minimal fluctuation width, 4-dimensional lattice, 4-dimensional hyperball, internal state capacity, volume gap, splitting representation, duality breaking, central projection, curved state space, self-similar hierarchy, toy model

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1. Introduction

In the preceding Paper 1, we placed between wavelength space and frequency space the reciprocal dual condition

$$\lambda_n = \frac{1}{\nu_n},$$

and observed that, when a constant sum-of-squares condition is further imposed on the wavelength space and the frequency space, a system of 5 components with 1 constraint retains 4 degrees of freedom. We also confirmed that, in the logarithmic variables

$$q_n = \log \lambda_n, \qquad p_n = \log \nu_n,$$

the reciprocal dual condition is expressed as the sign-reversal symmetry

$$p_n = -q_n.$$

In the preceding Paper 2, we formulated the unit-cell counting on the 4-dimensional integer lattice and computed the number $N_0(R)$ of unit cells fully inscribed in a 4-dimensional hyperball of radius R. In particular, we confirmed that the values

$$N_0(1) = 1, \qquad N_0(2) = 9, \qquad N_0(3) = 137$$

are obtained.

In the preceding Paper 3, we provided a supplement connecting this 4-dimensional lattice counting with the closed 4-dimensional structure and with central-projection-like observation.

In this paper, we reconstruct these three observations from smaller assumptions. The central assumption is the componentwise reciprocal duality

$$\lambda_n \nu_n = 1.$$

In this paper we read this condition not as a mere algebraic relation, but as the minimal dual structure placed between wavelength space and frequency space.

The question of this paper can be stated as follows.

> When a minimal conjugate width, 4-dimensional cell counting, and the composite radius $R^2 = \sum_n \nu_n^2$ are added to the minimal dual condition $\nu\lambda = 1$, what internal state structure, volume gap, splitting representation, and duality breaking are observed?

In this paper, we treat these not as a physical theory but as an observational model. Therefore, terms in this paper such as "energy-like," "ground," "quasi-ground," "splitting," "vacuum," and "symmetry breaking" are not identified with the concepts of the same names in standard physics. They are restricted terms for describing the geometric behavior of wavelength-frequency dual cells.

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2. Related Work and Positioning of This Paper

The problem awareness of this paper has, in part of its problem setting, associative points of contact with the body of research that does not regard spacetime as a background given from the start, but considers the possibility that spacetime and geometry emerge from more fundamental discrete structures, relational structures, quantum states, or pregeometric structures.

In causal set theory, instead of a spacetime continuum, a locally finite partially ordered set is taken as the foundation, with the order relation treated as primitive causality and local finiteness as discreteness [Surya 2019]. In Quantum Graphity, the possibility is examined that in the high-energy phase a highly connected graph state with permutation symmetry exists, while in the low-energy phase the symmetry is broken and low-dimensional, local structure appears [Konopka and Markopoulou 2006; Konopka, Markopoulou and Severini 2008]. In Group Field Theory condensate cosmology, the possibility is discussed that continuous cosmological dynamics emerges from condensate states of non-spatiotemporal quantum building blocks [Oriti 2021; Pithis and Sakellariadou 2019]. In the context of pregeometry, the possibility is also discussed that the difference between time and space is not built in from the start but appears as a result of spontaneous symmetry breaking [Wetterich 2021]. The standpoint that reads the quantum vacuum as a condensed-matter-like medium is also instructive for thinking of the vacuum not as a structureless void but as a state with internal degrees of freedom [Volovik 2003].

However, this paper does not extend, modify, or unify these existing theories. The structure of this paper is more restricted. Namely, starting from the reciprocal dual condition

$$\lambda_n \nu_n = 1,$$

we investigate what observational structures appear when a minimal conjugate width, 4-dimensional lattice cell counting, the volume gap, and the possibility of decomposition into finite-radius cells are added to it.

Therefore, the positioning of this paper is not that of a candidate for existing quantum gravity theories or unified theories, but a **minimal toy observation model based on wavelength-frequency duality**.

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3. Basic Assumptions

3.1 Reciprocal dual condition

In this paper, for each component we impose

$$\lambda_n \nu_n = 1,$$

where λ_n is a component on the wavelength-space side and ν_n is a component on the frequency-space side.

In this paper, we do not identify λ_n and ν_n with spatial coordinates, temporal frequencies, energy, momentum, or other quantities in standard physics. They are two dual variables in mutual reciprocal relation.

3.2 Composite frequency radius

On the frequency-space side, we define the composite radius

$$R^2 = \sum_{n=1}^{d} \nu_n^2,$$

where d is the number of frequency components; in this paper we treat $d = 4$, corresponding to the 4 degrees of freedom left by the 5-component, 1-constraint system of the preceding Paper 1 (§4.1).

In this paper, we treat R not as a substantive radius given from outside, but as a composite quantity determined from the sum of squares of the internal frequency components. In this sense, R can be read as an energy-like scale, or as a model label expressing the size of the internal frequency state.

However, in this paper we do not identify R with physical energy, mass, curvature radius, cosmic radius, or the like.

3.3 Minimal conjugate width

Ideally,

$$\lambda_n \nu_n = 1,$$

but if an observed state is treated as a perfect mathematical point, there is no room left to treat 4-dimensional lattice cell counting or splitting representations. Therefore, in this paper we assume the possible existence of a minimal conjugate width:

$$\delta^2 \geq \delta_{\min}^2,$$

where δ is a model quantity expressing the observational or constitutive width possessed by a wavelength-frequency dual cell. In this paper we do not derive the value of $\delta_{\min}$. Where needed, we use $\delta_{\min}^2 = 1/2$ as an illustrative value, but this is not a physical constant; it is a placeholder to indicate the scale of the discussion.

3.4 States are conjugate cells, not points

From the above, in this paper a state is treated not as a point

$$(\lambda, \nu)$$

but as a conjugate cell centered on the reciprocal dual condition and possessing the minimal width $\delta_{\min}$.

Therefore, the $R = 1$ state is also not a single point

$$\nu = 1, \qquad \lambda = 1,$$

but the minimal conjugate cell centered at $\nu = 1, \lambda = 1$.

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4. State Capacity at $R = 1, 2, 3$

4.1 Fully-inscribed cell count in 4 dimensions

In the preceding Papers 1 and 3, we confirmed that the 5-component, 1-constraint system can be read as a closed structure with 4 degrees of freedom. In this section, as the counting region corresponding to those 4 degrees of freedom, we consider the same 4-dimensional integer lattice as in the preceding Paper 2:

$$k = (k_1, k_2, k_3, k_4) \in \mathbb{Z}^4.$$

A 4-dimensional unit cell of side 1 is associated with each lattice point k. The half-width of this unit cell along each axis is $1/2$; this does not derive the value of $\delta_{\min}$ of the previous section, but means that we use the unit-cell normalization of Paper 2. The condition for this unit cell to be fully inscribed in a 4-dimensional hyperball of radius R is

$$\sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq R^2.$$

Therefore, we define the fully-inscribed cell count as

$$N_0(R) = \# \left\{ k \in \mathbb{Z}^4 \mid \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq R^2 \right\}.$$

Here $N_0(R)$ is not a particle number. Nor is it the number of material cells filling real space. $N_0(R)$ is the geometric capacity for placing unit cells without overlap inside the model state space of radius R.

4.2 State capacity at small radii

Based on the counting of the preceding Paper 2, at integer radii $R = 1, 2, 3$ we obtain

$$N_0(1) = 1,$$

$$N_0(2) = 9,$$

$$N_0(3) = 137.$$

Therefore the shell differences are

$$\Delta N_0(2) = N_0(2) - N_0(1) = 8,$$

$$\Delta N_0(3) = N_0(3) - N_0(2) = 128.$$

In the toy model of this paper, this sequence can be read as follows.

- $R = 1$ is a ground-like state corresponding to one minimal conjugate cell.
- $R = 2$ is a state in which, in addition to the central cell, first-neighbor cells appear in the 8 positive and negative directions of the 4-dimensional lattice.
- $R = 3$ is a state in which 128 further cells are added, and a large internal state shell appears that goes beyond the simple first-neighbor shell.

Figure 1. Exact sections of the $R = 3$ four-dimensional cell model. We first enumerate the 137 four-dimensional hypercubes of side 1 fully inscribed in the 4-dimensional hyperball B_3^4 of radius $R = 3$; cutting this cell set at $t = 0$ yields 57 three-dimensional cubic sections. Cutting further at $z = 0$ yields 21 unit-square sections on the central two-dimensional plane. The two-dimensional diagram at the right end is not a schematic shell diagram but the central section actually obtained from the set of 137 four-dimensional cells.

4.3 State capacity table

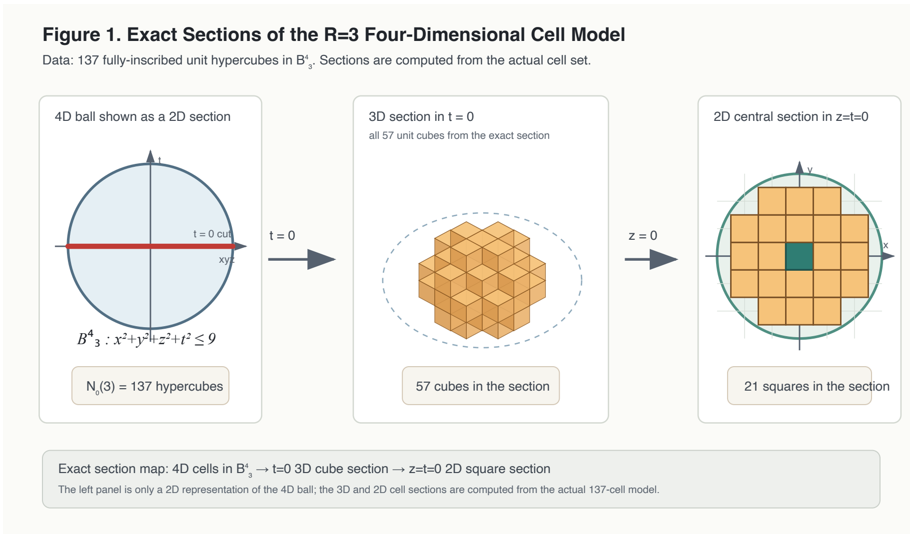

Figure 1: Figure 1. Exact Sections of the R=3 Four-Dimensional Cell Model

R	$N_0(R)$	Shell difference $\Delta N_0(R) = N_0(R) - N_0(R-1)$	Reading
1	1	1	Minimal conjugate cell, ground-like state
2	9	8	First-neighbor shell, 8 quasi-ground directional states
3	137	128	Large internal state shell, 128 additional states
4	473	336	Higher shell including composite directions
5	1,545	1,072	Growth of internal state capacity

R	$N_0(R)$	Shell difference $\Delta N_0(R) = N_0(R) - N_0(R-1)$	Reading
6	3,457	1,912	Higher internal state capacity
7	7,281	3,824	Higher internal state capacity
8	12,833	5,552	Higher internal state capacity
9	22,409	9,576	Higher internal state capacity
10	34,569	12,160	Higher internal state capacity

What this table shows is that the internal state capacity grows rapidly as R increases. However, rather than interpreting each state as superposed on the same physical point, we read them as unit cells with distinct position labels or phase labels in the model state space.

4.4 Classical reading and state-theoretic reading

Classically, the 8 states added at $R = 2$ can be read as minimal orbit candidates appearing around the central state. On the 4-dimensional lattice, these correspond to the positive and negative directions of each of the 4 axes.

State-theoretically, the same 8 can be read as quasi-ground states of the first shell adjacent to the ground cell.

The 128 states added at $R = 3$ can be read as a large internal state shell that includes not only single-axis directions but also combinatorial states spanning multiple axes. However, in this paper we do not identify them with multiplets of physical particles, spin, gauge degrees of freedom, particle species of the Standard Model, or the like.

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5. Volume Gap and Consistency

5.1 4-dimensional hyperball volume

The volume of the 4-dimensional hyperball is

$$V_4(R) = \frac{\pi^2}{2} R^4.$$

On the other hand, since the fully-inscribed cells are 4-dimensional unit cells of side 1, the stacked cell volume is

$$V_{\text{cell}}(R) = N_0(R).$$

Therefore, we define the difference between the continuous 4-dimensional hyperball volume and the discrete cell volume as

$$\Delta V(R) = V_4(R) - N_0(R),$$

and define the filling fraction as

$$\eta(R) = \frac{N_0(R)}{V_4(R)}.$$

5.2 Volume comparison table

R	$N_0(R)$	$V_4(R) = \frac{\pi^2}{2} R^4$	$\Delta V(R) = V_4(R) - N_0(R)$	$\eta(R) = N_0(R)/V_4(R)$
1	1	4.9348	3.9348	0.2026
2	9	78.9568	69.9568	0.1140
3	137	399.7190	262.7190	0.3427
4	473	1263.3094	790.3094	0.3744
5	1,545	3084.2514	1539.2514	0.5009
6	3,457	6395.5037	2938.5037	0.5405
7	7,281	11848.4601	4567.4601	0.6145
8	12,833	20212.9498	7379.9498	0.6349
9	22,409	32377.2372	9968.2372	0.6921
10	34,569	49348.0220	14779.0220	0.7005

From this table, two observations are obtained.

First, the filling fraction $\eta(R)$ shows local increases and decreases, but in the radius sweep of the preceding Paper 2 it shows a tendency to approach the continuous 4-dimensional hyperball volume at sufficiently large R.

Second, the volume gap $\Delta V(R)$ as an absolute quantity grows. This is because the unfilled region remaining in the fully-inscribed cell counting corresponds to the boundary layer of the 4-dimensional hyperball, and its dominant scale is given by the surface area of the 4-dimensional ball, that is, the R^3 order. Therefore, while the system appears to approach the continuum in relative terms, the absolute amount of mismatch remains as a boundary-layer scale.

This observation is important. For if one reads a state as a single state with high R, the internal state capacity becomes large, but at the same time the difference between the continuous hyperball volume and the discrete cell volume also becomes large.

5.3 On the boundary layer and asymptotic order

The $\Delta V(R)$ defined in this paper is the direct difference between the volume of the 4-dimensional hyperball of the same radius R and the fully-inscribed cell volume. This direct difference includes the unfilled layer near the boundary. The boundary-layer volume of a 4-dimensional ball of radius R, if the thickness is fixed at about the unit-cell width, is proportional to the surface-area scale of the 4-dimensional ball. The surface area of the 4-dimensional ball is

$$\frac{d}{dR}V_4(R) = \frac{d}{dR}\left(\frac{\pi^2}{2}R^4\right) = 2\pi^2 R^3,$$

so the dominant scale of the volume gap is of order R^3.

In this paper, we do not treat the asymptotic constant of $\Delta V(R)$ or a detailed evaluation of lattice fluctuations. The observation needed here is that, even as the filling fraction increases, the volume gap as an absolute quantity remains as a boundary-layer-derived quantity of order R^3.

5.4 Reading the volume gap as a mismatch indicator

In this paper, we tentatively read $\Delta V(R)$ as a mismatch quantity, or as an indicator for measuring geometric consistency.

In this reading, as R becomes larger the state capacity increases, but at the same time the mismatch quantity viewed as a single high-R state also increases. Therefore, as far as this indicator alone is concerned, there are cases where a representation decomposed into many low-R' states has a smaller mismatch quantity than a single state with high R.

However, this consistency is not identified with thermodynamic stability, quantum mechanical stability, or vacuum stability in field theory. It is strictly a geometric comparison indicator within this model.

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6. Splitting into Finite-R' States

6.1 R^2 as the conserved quantity

In this paper, as the quantity conserved before and after splitting, we consider not R but R^2. That is, when a single state R is decomposed into multiple finite states R'_a, we assume

$$R^2 = \sum_a {R'_a}^2.$$

This is because R^2 is defined as

$$R^2 = \sum_n \nu_n^2.$$

That is, R^2 is the sum of squares of frequency components, and it is natural to read it as a distributable energy-like quantity.

6.2 Condition favoring splitting

Let the mismatch quantity of the single state be $\Delta V(R)$, and let the mismatch quantity after splitting be

$$\sum_a \Delta V(R'_a).$$

Then, if

$$\Delta V(R) > \sum_a \Delta V(R'_a),$$

the mismatch quantity after splitting is smaller from the viewpoint of the volume gap.

For example, considering the single state $R = 2$,

$$\Delta V(2) = \frac{\pi^2}{2} 2^4 - 9 \approx 69.9568.$$

On the other hand, splitting into four $R' = 1$ states as $2^2 = 1^2 + 1^2 + 1^2 + 1^2$ gives

$$4\Delta V(1) = 4\left(\frac{\pi^2}{2} - 1\right) \approx 15.7392.$$

Under this simple volume-gap indicator, the representation split into four $R' = 1$ states has a smaller mismatch quantity than the single $R = 2$ state.

Similarly, for $R = 3$,

$$\Delta V(3) \approx 262.7190,$$

while splitting into nine $R' = 1$ states gives

$$9\Delta V(1) \approx 35.4132.$$

Therefore, as far as the simple indicator of volume-gap minimization is concerned, a high-R state appears more consistent when decomposed into multiple low-R' states.

6.3 Effect of the minimal conjugate width

If a minimum value of δ^2 exists, states do not collapse to perfect points. A high-R state contains many internal cells, but since each cell has the minimal conjugate width, it becomes difficult to make the whole perfectly consistent as a single state.

For this reason, there are cases where a representation decomposed into a family of finite, small R' states satisfies the local conjugate conditions more easily than a single high-R state.

This observation does not derive decay, phase transitions, particle creation, or vacuum decay in standard physics. However, as a toy model, it exhibits a structure in which a family of local ground-like states of finite R' appears out of an undifferentiated high-R state.

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7. Global $R = \infty$ and Finite-R' Internal Structure

7.1 Not a single infinite-radius state

In this model, viewing the system globally, one can consider a limit such as

$$R_{\text{global}} \to \infty.$$

In this limit, externally or globally, the state space appears nearly flat and continuous.

However, by the volume-gap comparison of the previous section, reading it as a single gigantic R state entails a large mismatch quantity. Therefore, it is more natural to read the global $R = \infty$ state not as a single infinite-radius cell, but as a collection of many finite R'_a states.

That is,

$$R_{\text{global}}^2 = \sum_a {R'_a}^2,$$

where each R'_a is a local conjugate cell with a finite, often small, value.

7.2 A collection of finite cells as a vacuum-like state

In this reading, the vacuum-like state of this paper is not a structureless void. Globally it appears flat or of infinite radius, but internally it can be read as a family of conjugate cells of finite R'.

However, in this paper we do not identify this structure with the cosmological vacuum or the physical vacuum. The "vacuum-like state" here refers to a ground-like state space possessing internal state capacity, prior to appearing as local excitations in the observation space.

7.3 Filling of state space and sparseness of observation space

A point to note here is that $N_0(R)$ does not represent the matter density of real space. $N_0(R)$ is the number of placeable minimal conjugate cells in the wavelength-frequency dual state space.

Therefore, even if many finite-R' states exist, this does not mean that the observation space is filled without gaps by material cells. Rather, what appears in the observation space is not these ground cells themselves, but the extension projected through relative phase mismatches, boundary gaps, local excitations, or duality breaking.

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8. ν-λ Duality Breaking

8.1 Undifferentiated dual phase

The minimal assumption of this paper is

$$\lambda\nu = 1.$$

At this stage, one can also read that which variable is called the "wavelength" and which the "frequency" is not yet fully distinguished.

That is, in the initial state there is only the reciprocal relation between the two dual variables, and the differentiation of roles — one bearing internal state density and the other bearing extension — has not yet arisen.

8.2 Role differentiation through splitting

Considering the splitting into finite-R' cells, on the frequency side many internal states are arranged densely. On the other hand, by reciprocal duality, on the wavelength side the corresponding states appear as spread or extension.

Therefore, even when looking at the same state, on the

$$\nu \text{ side}$$

it appears as dense internal state filling, while on the

$$\lambda \text{ side}$$

it appears as a sparse extension.

That is,

$$\text{state density on the } \nu \text{ side} \neq \text{spatial density on the } \lambda \text{ side.}$$

Owing to this asymmetry, the state space appears dense when viewed from the frequency side, but appears sparse when viewed from the wavelength side.

8.3 Restriction concerning the relation to time and space

ν, as a frequency, can be read as a quantity related to internal updating, period, and state density. On the other hand, λ, as a wavelength, can be read as a quantity related to extension, spread, and placeability.

For this reason, this model can be read as a toy model in which an “updating aspect” and an “extensional aspect” differentiate out of an undifferentiated dual state.

However, in this paper we do not identify ν with time, nor λ with space. Nor do we claim to have derived the origin of time and space. What this paper states is only that, when the reciprocal duality and the splitting representation are taken into account, a role differentiation of the dual variables is observed.

8.4 Relation to spontaneous symmetry breaking

The duality breaking referred to in this paper does not derive spontaneous symmetry breaking in the Standard Model. What is meant here is a geometric, state-theoretic role differentiation in which two variables that appeared equivalent in the undifferentiated dual phase come to bear different roles in the post-splitting state structure.

Therefore, as the expression of this paper, it is appropriate to call it not “spontaneous symmetry breaking itself” but a “role differentiation analogous to spontaneous symmetry breaking.”

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9. Central Projection, Curvature, and Hierarchical Structure

9.1 R is determined from internal frequencies

In this paper, since

$$R^2 = \sum_n \nu_n^2,$$

R does not exist first as an external background but is determined from the internal frequency components.

In this standpoint, what is more fundamental is not R but ν. R is composed from the decomposition structure of ν.

9.2 Curved state space

Connecting with the central-projection-like observation treated in the preceding Paper 3, R can be read as the radius or curvature scale of the state space. However, in this paper we do not identify this with the curvature of physical spacetime.

What is important is that this model does not place cells on a flat background space from the start; rather, it treats cell capacity, the volume gap, and splitting representations inside a curved state space defined by R.

Figure 2. Exact 3-dimensional inverse central projection from the tangent plane to the hemisphere. The 21 unit-square sections at the right end of Figure 1 are read as a cell set on the xy plane tangent to the north pole of the 3-dimensional hemisphere of radius $R = 3$. The line connecting each vertex $P = (u, v, 3)$ on the tangent plane with the center O intersects the hemisphere of radius 3 at exactly one point $Q = 3P/\|P\|$. In the figure, this map is applied to all 32 vertices obtained from the 21 squares. The reinterpretation into the curved state space referred to in this paper is schematized as the inverse map of central projection in this sense. However, this does not represent the curvature of physical spacetime.

9.3 Self-similar hierarchy

When a state of some radius R is decomposed into a family of states of finite radii R'_a, each R'_a can in turn have internal structure of the same form,

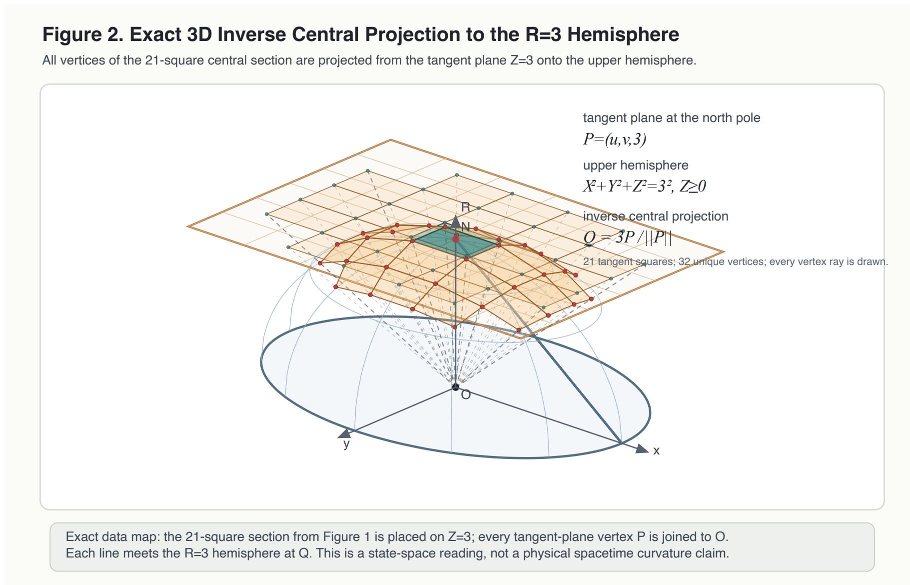

Figure 2: Figure 2. Exact 3D Inverse Central Projection to the R=3 Hemisphere

$$R_a'^{\,2} = \sum_i \nu_{a,i}^2.$$

Therefore, a hierarchy

$$R \longrightarrow \{R_a'\} \longrightarrow \{R_{a,b}''\} \longrightarrow \cdots$$

appears naturally.

We do not claim that this hierarchy is a figure with a strict fractal dimension. However, in the sense that the same dual condition and the same composite radius condition appear recursively at different scales, it has a self-similar, or fractal-like, structure.

Figure 3. Exact 3-dimensional projection of hierarchical curved state spaces. A global sphere of radius $R = 20$ is placed, and local spheres of radius $R' = 6$ are arranged centered at 4 points: its north pole, south pole, and the points at 0° and 180° on the equator. Furthermore, for each local sphere, a secondary sphere of radius $R'' = 2$ is placed centered at the surface point advanced from the center C outward along the normal direction by $6C/\|C\|$. Thus the center of each $R' = 6$ sphere lies on the $R = 20$ sphere, and the center of each $R'' = 2$ sphere lies on the corresponding $R' = 6$ sphere. This figure is a 3-dimensional projection of a toy model showing the hierarchical placement rule, and does not claim that physical spacetime is fractal.

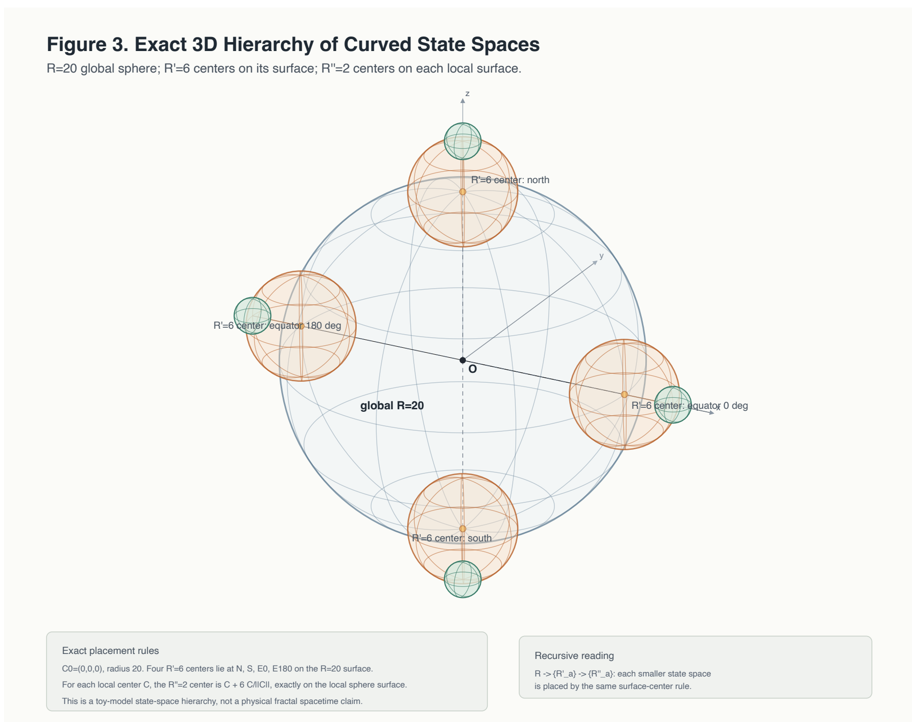

Figure 3: Figure 3. Hierarchical Curved State Spaces

9.4 Observation as a curved hierarchy

From the above, the state space of this model is not a simple lattice cell set placed on a flat background. Rather, it is observed as a hierarchical structure in which a curved state space contains smaller curved state spaces, which in turn have internal structure of the same form.

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10. Scope of Claims of This Paper

What this paper claims is limited to the following.

1. From the reciprocal dual condition $\lambda_n \nu_n = 1$, a dual structure of wavelength space and frequency space can be defined.
2. Assuming a minimal conjugate width $\delta_{\min}$, a state can be treated not as a point but as a minimal conjugate cell.
3. Via $R^2 = \sum_n \nu_n^2$, R can be read as a composite radius determined from the internal frequency components.
4. In the 4-dimensional fully-inscribed cell counting, $N_0(1) = 1$, $N_0(2) = 9$, $N_0(3) = 137$ appear.
5. $R = 1$ can be read as the minimal conjugate cell, $R = 2$ as 8 first-neighbor quasi-ground states, and $R = 3$ as a large internal state shell containing 128 additional states.
6. The difference $\Delta V(R)$ between the 4-dimensional hyperball volume $V_4(R)$ and the cell volume $N_0(R)$ corresponds to the boundary-layer volume and therefore grows mainly at order R^3.
7. Reading the volume gap as a mismatch quantity, there are cases where a high-R state appears more consistent when decomposed into a family of finite-R' states.
8. After splitting, the ν side appears to differentiate into internal state density and the λ side into extension and spread.
9. Connecting with central-projection-like observation, curved state spaces appear hierarchically and self-similarly.

What this paper does not claim is as follows.

1. This paper does not modify standard physics.
2. This paper does not derive general relativity, quantum theory, quantum field theory, or the Standard Model.
3. This paper does not prove the origin of time or space.
4. This paper does not derive physical constants, the elementary particle spectrum, interactions, gauge fields, or gravity.
5. This paper does not identify 137 with the fine-structure constant.
6. This paper does not derive the value of $\delta_{\min}$.
7. This paper does not give a rigorous asymptotic expansion of the volume gap.
8. This paper does not assert cosmological reality.

11. Discussion

11.1 Why complex structure appears from minimal hypotheses

The assumptions of this paper are very few. The central one is

$$\lambda\nu = 1,$$

to which only the minimal conjugate width and the 4-dimensional cell counting are added.

Nevertheless, the following structures appear in a chained manner:

- the minimal conjugate cell
- internal state capacity
- shell structure
- the volume gap
- consistency in splitting representations
- the duality of global infinite radius and local finite radius
- role differentiation between the ν side and the λ side
- curved hierarchical structure
- self-similar decomposition

This shows that the reciprocal dual condition is not a mere algebraic formula, but carries very strong constraints as a constitutive principle of state space.

11.2 Why it looks sparse

Inside the state space, many finite-R' cells can exist. Viewed from the frequency side, this appears as dense internal states.

However, on the wavelength side, by reciprocal duality they appear as spread. Therefore, the same structure looks packed on the ν side and looks sparse on the λ side.

This observation provides a toy model that reads the vacuum not as "empty space with nothing in it" but as "a state in which internal states exist but appear on the wavelength side as a sparse extension."

11.3 Why this paper must be written cautiously

The structure of this paper evokes associations with the differentiation of time and space, vacuum structure, spacetime emergence, self-similar cosmologies, and so on. However, asserting these directly would exceed the verifiable scope of the model.

For this reason, this paper states only the observational structures that appear from the minimal reciprocal dual condition. Room is left for readers to examine possible physical correspondences, but this paper itself makes no physical identification.

This caution is not a weakness of the model, but a condition for enhancing its reusability.

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12. Conclusion

In this paper, starting from the minimal reciprocal dual condition placed between wavelength components λ_n and frequency components ν_n,

$$\lambda_n \nu_n = 1,$$

we organized the minimal conjugate width, the 4-dimensional fully-inscribed cell counting, the volume gap, the finite-R' decomposition, duality breaking, and the curved hierarchical structure as a single observational model.

The main conclusions are as follows.

First, a state can be treated not as a point but as a minimal conjugate cell centered at $\nu = 1, \lambda = 1$. $R = 1$ can be read as the ground-like state corresponding to this minimal conjugate cell.

Second, at $R = 2$ eight first-neighbor quasi-ground states appear, and at $R = 3$ 128 additional states appear. This shows that the internal state capacity grows rapidly with R.

Third, by comparing the 4-dimensional hyperball volume with the fully-inscribed cell volume, the filling fraction shows an asymptotically increasing tendency accompanied by local fluctuations, while the volume gap as an absolute quantity remains as a boundary-layer-derived quantity of order R^3. Reading this volume gap as a mismatch quantity, there are cases where a high-R state appears more consistent when decomposed into a family of finite, small R' states.

Fourth, the post-splitting states are observed as dense internal states on the ν side and as a sparse extension on the λ side. This can be read as a role differentiation of the wavelength-frequency duality, that is, a structure analogous to spontaneous symmetry breaking.

Fifth, R is not a radius given from outside but a composite quantity determined from the internal frequency components. Therefore, an R state can be decomposed into finite-R' states, and each R' state can in turn have internal structure of the same form. In this sense, this model has a curved, self-similar hierarchical structure.

This paper does not claim a unified physical theory. However, it is a reusable toy observation model showing that, from the minimal assumption $\nu\lambda = 1$, the internal state capacity, the volume gap, splitting representations, duality breaking, and curved hierarchical structure can be read in a chained manner.

Appendix A: Definition of the Fully-Inscribed Cell Counting

Consider the unit cell of side 1 associated with the 4-dimensional lattice point

$$k = (k_1, k_2, k_3, k_4) \in \mathbb{Z}^4.$$

The half-width in each direction is 1/2. For the whole unit cell to be fully inscribed in the 4-dimensional hyperball of radius R, it suffices that the farthest vertex of the cell lies within radius R.

Therefore the condition is

$$\sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq R^2.$$

The fully-inscribed cell count is defined by

$$N_0(R) = \# \left\{ k \in \mathbb{Z}^4 \mid \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq R^2 \right\}.$$

Appendix B: Python Code for Reproduction

The following is minimal code that computes $N_0(R)$, the 4-dimensional hyperball volume, the volume gap, and the filling fraction for integer radii $R = 1$ to $R = 10$.

```python
import math
import itertools

def N0(R):
    M = int(math.ceil(R))
    count = 0
    for k in itertools.product(range(-M, M + 1), repeat=4):
        s = sum((abs(x) + 0.5)**2 for x in k)
        if s <= R*R + 1e-12:
            count += 1
    return count
```

```
prev = 0
for R in range(1, 11):
    n = N0(R)
    shell = n - prev
    V4 = (math.pi**2 / 2.0) * R**4
    gap = V4 - n
    eta = n / V4
    print(R, n, shell, V4, gap, eta)
    prev = n
```

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